EEG/ERP Processing Pipeline Summary: Ramzaoui & Beck (2025)

This document details the complete pipeline for the Ramzaoui & Beck (2025) study, organized by its folder structure and processing phases. The pipeline converts raw data into quantified metrics (N2pc, CDA, N2, FN400) and includes two key exploratory time-domain (Negative Slow Wave) and time-frequency (Alpha) analyses.

Top-Level Directory Structure 📁

The main study directory contains the following key folders and files:

- **EEG_ERP_Processing:** Main folder housing the core Matlab Scripts 1-11 (Phase I–IV). This folder also contains essential TXT and XLSX files (e.g., IC removal lists, channel operations, bin definitions, and re-referencing lists) necessary for running certain scripts.
- **ERP_Cluster_Analyses:** Folder containing the Matlab Scripts for the permutation test used to define ERP statistical time windows (Phase IV).
- **Exploratory_Analyses:** Folder housing the Negative Slow Wave and Alpha Matlab scripts (Phase V).
- **Data_Analysis_Scripts_R:** Folder housing the R statistical scripts (Phase VI).
- **Individual Subject Folders** (e.g., 76, 74): Each contains all raw and intermediate processed data files, including the figures.
- **BIDS_RAW_Dataset:** Folder containing the original raw data in BIDS format and associated metadata.
- **Get_BDF_Timestamps:** Folder containing one script enabling the extraction of BDF timestamps.
- **Script0:** Folder containing scripts to convert BDF data into the original dataset structure used for preprocessing and analyses. Note: This conversion is already complete, and the output files (e.g., 1.set, 2.set) are present in the subject folders. EEG preprocessing can directly start with Script 1.

Three files are present in the main directory for data integrity and event management:

- **Laboratory_log_VWM-PerceptualGrouping_2024-2025.pdf:** Contains essential project metadata including the title, PIs, data collection dates, location, and key equipment used (EEG Biosemi 64 electrodes system).
- **allERPs_Subject_Rejection_List.xlsx:** Contains parameters/flags used for subject-specific exclusion criteria and artifacts parameters (Phase II).
- **allERPs_Participant_Demographics.xlsx:** Contains demographic data of the final sample size N=53 participants (Phase VI).
- **allERPs_Event_Code_Scheme.xlsx:** Defines the event codes (Phase II).

Phase I: Pre-processing & Artifact Correction

1. **Script 1: Initial Processing and Filtering**
    - Data Loading & Downsampling: Loads EEG data and reduces the sampling rate (e.g., to 256 Hz).
    - Filtering: Applies a mild high-pass filter (0.01 Hz) and remove DC offsets.
    - Channel location: Add channel location information corresponding to the 3-D coordinates of the electrodes based on 10-10 International System site locations.
    - Re-referencing & Bipolar Channels: Re-references the data to the Left Mastoid (Lm) and creates uncorrected bipolar channels (VEOG, HEOG).
2. **Scripts 2 & 3: ICA Optimization and Calculation**
    - Script 2 (Preparation): Further downsamples the data (to 100 Hz) and applies a stronger high-pass filter (1.0 Hz) to optimize the dataset for ICA. Delete segment of breaks between blocks and remove continous segments containing huge artifacts.
    - Script 3 (Calculation): Calculates the ICA weights and sphere matrices on the optimized dataset.
3. **Script 4: ICA Correction and Channel Interpolation**
    - ICA Correction: The ICA solution is applied to the full 256 Hz data. Manually identified artifactual Independent Components (ICs) are subtracted from the EEG data.
    - Final Re-referencing: The data is re-referenced to the Average of the Left and Right Mastoids.
    - Interpolation: Bad channels are corrected using spherical spline interpolation.

Phase II: Epoching & Artifact Rejection

4. **Script 5: Event Segmentation and Epoching**
    - Event Handling: Creates an event list and uses Bin Descriptor Files (BDFs) to assign events to specific analysis bins.
    - Epoching: Segments the data into ERP epochs:
        - Stimulus Epochs: Baseline corrected, typically -200 ms to 1000 ms.
        - Probe Epochs: Baseline corrected, typically -200 ms to 800 ms.
5. **Script 6: Epoch-Specific Artifact Rejection**
    - Artifact Detection: Applies subject- and ERP-specific parameters (detailed in external CSV/Excel files) for:
        - Blink/Saccade Detection (Step Function).
        - CRAP Detection (Simple Voltage Threshold - SVT). CRAP: Commonly Recorded Artifactual Potentials.
        - CRAP Detection (Moving Window Peak-to-Peak Amplitude - MW).
    - Output: Three specialized artifact-rejected datasets are saved (6a: Stimulus locked N2pc/CDA, 6b: Probe locked N2pc, 6c: Probe locked N2/FN400).

Phase III: Individual Subject ERPs & Group Prep

6. **Script 7: ERP Averaging, Difference Waves, and Final Filtering**
    - ERP Averaging: Creates averaged ERP waveforms (ERPsets).
    - Difference Waves: Calculates key difference waveforms using bin operators:
        - Contra minus Ipsi (7a: Stimulus locked N2pc, CDA).
        - Contra minus Ipsi of Change minus No-Change (7b: Probe locked N2pc).
        - Change minus No-Change (7c: Probe locked N2, FN400).
    - Channel Averaging: Averages data across specific Clusters of Interest (COI).
    - Final Filtering: Applies a non-causal 20 Hz low-pass filter to all averaged waveforms for visualization.
7. **Script 8: Individual Subject ERP Plotting**
    - Stimulus Plots (8a): Plots the N2pc/CDA Contra minus Ipsi difference waves at the COI, and the HEOG difference wave.
    - Probe N2pc Plots (8b): Plots the Probe N2pc Contra minus Ipsi of Change minus No-Change difference waves at the COI, and the HEOG waveform.
    - Probe N2/FN400 Plots (8c): Plots the N2/FN400 Change minus No-Change difference waves at fronto-central sites.

Phase IV: Group Analysis & Quantification

8. **Script 9: Grand Average Calculation**
    - Grand Average (GA): Loads individual ERPsets and computes the Grand Average waveform for every analysis bin and difference wave (9a: Stimulus locked N2pc/CDA, 9b: Probe locked N2pc, 9c: Probe locked N2/FN400).
    - SEM Calculation: Includes the Standard Error of the Mean (SEM) necessary for plotting and statistical visualization.
9. **Script 10: Grand Average Plotting**
    - GA Plotting: Loads the filtered GA ERPsets from Script 9 and plots the difference waves and cluster averages (10a: Stimulus locked N2pc/CDA, 10b: Probe locked N2pc, 10c: Probe locked N2/FN400).
    - Visualization: Generates PDF figures, often including shaded SEM areas.
10. **Permutation Test (folder)**
    - Data-Driven Window Definition: Uses the **clustGND** function to perform temporal cluster analysis.
    - Output: Statistically significant time windows are determined for each ERP.
11. **Script 11: ERP Measurement**
    - ERP Measurement: Uses **pop_geterpvalues** to extract key metrics (e.g., Mean Amplitude and Latency).
    - Window Use: Utilizes the specific time windows defined by the Permutation Test.
    - Output: Final text files (.txt) containing all subject-level Mean Amplitudes and Latencies ready for ANOVA or other statistical testing. The data are exported into the ERP_Measurements folder, ready for analysis in R.

Phase V: Exploratory Analyses (NSW & Alpha)

This phase contains two distinct exploratory time-frequency and ERP analyses: **Negative Slow Wave (NSW)** and **Alpha Oscillations**.

1. Negative Slow Wave (NSW) Analysis

This analysis focuses on a specific posterior component associated with sustained maintenance. These scripts utilize the data generated in Phase III.

- **Step1_Compute_Negative_Slow_Wave.m**
  - **Purpose:**Computes the NSW waveform by performing ROI averaging over posterior channels and trial-level baseline correction, saving both filtered and unfiltered waveforms.
- **Step2_Permutation_Test_Negative_Slow_Wave.m**
  - **Purpose:** Performs the temporal cluster permutation test on the NSW to determine statistically significant time windows.
- **Step3_Plots_Negative_Slow_Wave.m**
  - **Purpose:** Generates the final figures for the analysis, including the Grand Average waveform and mean amplitude bar charts using the time windows found in the permutation test.

2. Alpha Oscillations Analysis (Time-Frequency)

This analysis focuses on alpha power, a measure of working memory load and maintenance.

- **Step1_Compute_Alpha_Power.m**
  - **Purpose:** Calculates the 8-12 Hz alpha power time series using the Hilbert transform and computes the % change from baseline.
- **Step2_Plots_Alpha_Power.m**
  - **Purpose:** Generates the time-resolved power (% change) plot and mean amplitude bar charts.
- **Step3_Bays_Factor_Alpha_Power.m**
  - **Purpose:** Performs the paired t-tests (with Bonferroni correction) and exports the subject data into CSV format for subsequent Bayesian t-tests in R.

Phase VI: STATISTICAL ANALYSES CONDUCTED IN R

The final phase uses the ERP metrics (from Phase IV), Alpha Power data (from Phase V), and Behavioral Data, primarily focusing on ANOVAs, t-tests and Bayesian t-tests.

Behavioral Data Modeling

- **#Step1a_Models_1_7_Accuracy.R**
  - Runs the analyses on the accuracy data.

- **#Step1b_Exploratory_Dprime.R**
    - Runs exploratory analyses on the d' (D-prime) sensitivity index.

ERP Metric Analyses (N2pc, CDA, N2, FN400)

- **#Step2_Models_2_3_N2pc_CDA_Mean_Amplitude_StimulusType.R**
    - Runs analyses on the mean amplitude for N2pc and CDA by stimulus type.
- **#Step3_Models_4b_4c_N2_FN400_Onset_Latency_ProbeType_plusExploratory_byStimulusType.R**
    - Runs analyses on N2 and FN400 onset latency by probe and stimulus type.
- **#Step4_Models_5a_6a_N2_FN400_Mean_Amplitude_StimulusType**
    - Runs analyses on the mean amplitude for N2 and FN400 by stimulus type.
- **# Step5_Models_5b_6b_N2_FN400_Peak_Latency_StimulusType**
    - Runs analyses on the peak latency for N2 and FN400 by stimulus type.
- **#Step6_Models_8b_9b_N2_FN400_Peak_Latency_Probetye.R**
    - Runs analyses on the peak latency for N2 and FN400 by probe type.
- **# Step7_Models_8a_9a_N2_FN400_Mean_Amplitude_ProbeType**
    - Runs analyses on the mean amplitude for N2 and FN400 by probe type.

Alpha Power and Exploratory Bayesian Analysis

- **Alpha_Power_Bayesian_TTest_Exports_Matlab/ (Folder)**
    - Contains CSV files from Matlab analyses with the paired subject data for Mean Alpha Power (% change) for Bayesian analysis.
- **#Step8_Exploratory_Bayesfactor_TTest_Alpha_Power.R**
    - Performs the Bayesian Paired t-tests and calculates the Bayes Factor (BF01) for alpha suppression effects.